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Inventor(s) Chuang; Harry-Hak-Lay et al.

Method for forming an integrated chip (IC) including forming a through substrate via (TSV) in an isolation structure formed in a first opening that formed in a metal substrate layer

Abstract

Various embodiments of the present disclosure are directed towards an integrated chip (IC). The IC includes a substrate. The substrate includes a metal layer, a device layer disposed over the metal layer, and an insulating layer disposed vertically between the metal layer and the device layer. A semiconductor device is disposed on the device layer. An interlayer dielectric (ILD) layer is disposed over the semiconductor device and the substrate.

Inventors: **Chuang; Harry-Hak-Lay (Zhubei, TW), Lin; Hsin Fu (Hsinchu, TW), Liu; Chien Hung (Hsinchu County, TW)**

Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.** (Hsinchu, TW)

Family ID: **83697851**

Assignee: **Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)**

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Primary Examiner: Mai; Anh D

Attorney, Agent or Firm: Eschweiler & Potashnik, LLC

Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional of U.S. application Ser. No. 17/397,160, filed on Aug. 9, 2021, which claims the benefit of U.S. Provisional Application No. 63/214,846, filed on Jun. 25, 2021. The contents of the above-referenced patent applications are hereby incorporated by reference in their entirety.

BACKGROUND
(1) Semiconductor devices are electronic components that exploit electronic properties of semiconductor materials to affect electrons or their associated fields. A widely used type of semiconductor device is metal-oxide-semiconductor field-effect transistor (MOSFET). Semiconductor devices have traditionally been formed on bulk semiconductor substrates. In recent years, semiconductor-on-insulator (SOI) substrates have emerged as an alternative to bulk semiconductor substrates. Among other things, an SOI substrate leads to reduced parasitic capacitance, reduced leakage current, reduced latch up, and improved semiconductor device performance (e.g., lower power consumption and higher switching speed).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1** illustrates a cross-sectional view of some embodiments of an integrated chip (IC) comprising a low-cost semiconductor-on-insulator (SOI) structure.
(3) FIG. **2** illustrates a cross-sectional view of some more detailed embodiments of the IC of FIG. **1**.
(4) FIG. **3** illustrates a cross-sectional view of some other embodiments of the IC of FIG. **2**.
(5) FIGS. **4**A-**4**D illustrate bottom views of some embodiments of the IC of FIG. **3**.
(6) FIG. **5** illustrates a cross-sectional view of some other embodiments of the IC of FIG. **3**.
(7) FIG. **6** illustrates a cross-sectional view of some other embodiments of the IC of FIG. **5**.
(8) FIGS. **7-15** illustrate a series of cross-sectional views of some embodiments of a method for forming an IC comprising a low-cost SOI structure.
(9) FIG. **16** illustrates a flowchart of some embodiments of a method for forming an IC comprising a low-cost SOI structure.
DETAILED DESCRIPTION
(10) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature

over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(11) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(12) Some integrated chips (ICs) comprise a plurality of semiconductor devices (e.g., insulated gate field-effect transistors (IGFETs)) disposed over/within a semiconductor-on-insulator (SOI) substrate. Typically, the SOI substrate comprises an insulating layer vertically separating a first semiconductor layer from a second semiconductor layer. Generally, the typical SOI substrate is formed via an SOI bonding process (e.g., bonding the second semiconductor layer to the insulating layer). The SOI bonding process is expensive, which increases the cost to fabricate an IC with the SOI substrate. Thus, a low-cost alternative to the typical SOI substrate is desirable to reduce the cost to manufacture ICs that utilize the benefits of SOI substrates.

(13) Various embodiments of the present disclosure are directed toward an integrated chip (IC) comprising a low-cost SOI structure (e.g., low-cost SOI substrate). The low-cost SOI structure comprises a device layer, an insulating layer, and a metal layer. The device layer is disposed over both the insulating layer and the metal layer, and the insulating layer vertically separates the metal layer from the device layer. A semiconductor device (e.g., IGFET) is disposed on/over the device layer, over the insulating layer, and over the metal layer.

(14) The IC comprising the low-cost SOI structure may cost less to fabricate than an IC comprising a typical SOI substrate due to the low-cost SOI structure being formed without the expensive SOI bonding process. More specifically, the low-cost SOI structure is formed by forming the metal layer over the insulating layer, which costs less than the expensive SOI bonding process. As such, the low-cost SOI structure may provide a low-cost alternative to the typical SOI substrate.

(15) FIG. **1** illustrates a cross-sectional view **100** of some embodiments of an integrated chip (IC) comprising a low-cost semiconductor-on-insulator (SOI) structure **102**.

(16) As shown in the cross-sectional view **100** of FIG. **1**, the IC comprises the low-cost SOI structure **102** (e.g., a low-cost SOI substrate). The low-cost SOI structure **102** comprises a device layer **104**, an insulating layer **106**, and a metal layer **108**. The device layer **104** is disposed over the insulating layer **106** and the metal layer **108**. The insulating layer **106** is disposed vertically between the metal layer **108** and the device layer **104**. The insulating layer **106** vertically separates the metal layer **108** from the device layer **104**. In some embodiments, the low-cost SOI structure **102** is referred to as a substrate (and/or an SOI substrate).

(17) The device layer **104** is a semiconductor material. The semiconductor material may be or comprise, for example, silicon (Si), germanium (Ge), silicon-germanium (SiGe), gallium arsenide (GaAs), some other semiconductor material, or a combination of the foregoing. In some embodiments, the device layer **104** is silicon (Si). In further embodiments, the device layer **104** is monocrystalline silicon. In some embodiments, the device layer has a thickness between about 2 micrometers (μm) and about 15 μm. In yet further embodiments, an upper surface of the device layer **104** defines a front-side **102*f*** of the low-cost SOI structure **102**.

(18) The insulating layer **106** is configured to electrically isolate the device layer **104** from the

metal layer **108**. The insulating layer **106** may be or comprise, for example, a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), a high-k dielectric material (e.g., a dielectric material with a dielectric constant greater than about 3.9, such as, hafnium oxide (HfO), tantalum oxide (TaO), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), aluminum oxide (AlO), zirconium oxide (ZrO), or the like), an oxide (e.g., silicon dioxide (SiO.sub.2)), a nitride (e.g., silicon nitride (SiN)), an oxy-nitride (e.g., silicon oxy-nitride (SiON)), undoped silicate glass (USG), doped silicon dioxide (e.g., carbon doped silicon dioxide), borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), a spin-on glass (SOG), some other dielectric material, or a combination of the foregoing. In some embodiments, the insulating layer **106** has a thickness between about 0.5 μm and about 3 μm.
(19) The metal layer **108** is disposed below both the insulating layer **106** and the device layer **104**. The metal layer **108** may be or comprise, for example, copper (Cu), aluminum (Al), tungsten (W), gold (Au), silver (Ag), platinum (Pt), some other metal, or a combination of the foregoing. In some embodiments, the metal layer **108** has a thickness between about 1 μm and about 5 μm. In further embodiments, a lower surface of the metal layer **108** defines a back-side **102***b* of the low-cost SOI structure **102**. In some embodiments, the metal layer **108** contacts (e.g., directly contacts) the insulating layer **106**. In further embodiments, the insulating layer **106** contacts (e.g., directly contacts) the device layer **104**.
(20) A semiconductor device **110** (e.g., insulated gate field-effect transistors (IGFETs)) is disposed on/over the device layer **104**. For example, the semiconductor device **110** comprises a pair of source/drain regions **112**, a gate dielectric **114**, and a gate electrode **116**. The pair of source/drain regions **112** are regions of the device layer **104** having a first doping type (e.g., n-type).
(21) The gate dielectric **114** is disposed over the device layer **104** and between the source/drain regions of the pair of source/drain regions **112**. The gate electrode **116** overlies the gate dielectric **114**. In some embodiments, the gate dielectric **114** and the gate electrode **116** are collectively referred to as a gate stack. In some embodiments, the gate electrode **116** is or comprises polysilicon. In such embodiments, the gate dielectric **114** may be or comprise, for example, an oxide (e.g., silicon dioxide (SiO.sub.2)). In other embodiments, the gate electrode **116** may be or comprise a metal, such as aluminum (Al), copper (Cu), titanium (Ti), tantalum (Ta), tungsten (W), molybdenum (Mo), cobalt (Co), or the like. In such embodiments, the gate dielectric **114** may be or comprise a high-k dielectric material, such as hafnium oxide (HfO), tantalum oxide (TaO), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), aluminum oxide (AlO), zirconium oxide (ZrO), or the like.
(22) An interlayer dielectric (ILD) structure **118** is disposed over both the semiconductor device **110** and the low-cost SOI structure **102**. The ILD structure **118** comprises one or more stacked ILD layers, which may respectively comprise a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), an oxide (e.g., SiO.sub.2), a nitride (e.g., SiN), an oxy-nitride (e.g., SiON), undoped silicate glass (USG), doped silicon dioxide (e.g., carbon doped silicon dioxide), borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), a spin-on glass (SOG), or the like.
(23) An interconnect structure **120** (e.g., metal interconnect) is disposed in the ILD structure **118** and over the low-cost SOI structure **102**. The interconnect structure **120** comprises a plurality of conductive contacts **122** (e.g., metal contacts), a first plurality of conductive lines **124** (e.g., metal wires), and a plurality of conductive vias **126** (e.g., metal vias). The conductive contacts **122** extend through the ILD structure **118** to contact the pair of source/drain regions **112** and the gate electrode **116**. The first plurality of conductive lines **124** and the plurality of conductive vias **126** are disposed over the conductive contacts **122** and alternate back and forth from the conductive contacts **122** toward an upper surface of the ILD structure **118**. For clarity in the figures, only some of the plurality of conductive contacts **122**, some of the first plurality of conductive lines **124**, and

some of the plurality of conductive vias **126** are labeled in the figures.
(24) The interconnect structure **120** is embedded in the ILD structure **118** and is configured to provide electrical connections between various devices of the IC. In other words, the first plurality of conductive lines **124**, the plurality of conductive vias **126**, and the plurality of conductive contacts **122** are electrically coupled together in a predefined manner and are configured to provide electrical connections between the various devices of the IC. In some embodiments, the first plurality of conductive lines **124** and the plurality of conductive vias **126** may be or comprise, for example, copper (Cu), aluminum (Al), gold (Au), silver (Ag), platinum (Pt), or the like. In further embodiments, the plurality of conductive contacts **122** may be or comprise, for example, tungsten (W), copper (Cu), aluminum (Al), or the like.
(25) The IC comprising the low-cost SOI structure **102** may cost less to fabricate than an IC comprising a typical SOI substrate due to the low-cost SOI structure **102** being formed without an expensive SOI bonding process. More specifically, the low-cost SOI structure **102** is formed by forming the metal layer **108** over the insulating layer **106** (described in more detail hereinafter), which is less expensive than the SOI bonding process.
(26) Further, even though the low-cost SOI structure **102** costs less to fabricate than the typical SOI substrate, the low-cost SOI structure **102** may provide substantially similar performance as the typical SOI substrate. More specifically, because the substrate comprises the metal layer **108**, and because the insulating layer **106** vertically separates the metal layer **108** from the device layer **104**, the metal layer **108** may be biased (e.g., applying a specific current and/or voltage to the metal layer **108**), thereby improving the performance of the semiconductor device **110** (e.g., increased switching speed, reduced leakage current, etc.). As such, the low-cost SOI structure **102** provides substantially similar performance improvements as the typical SOI substrate while costing less to fabricate. Thus, the low-cost SOI structure **102** may be a low-cost alternative to the typical SOI substrate.
(27) FIG. **2** illustrates a cross-sectional view **200** of some more detailed embodiments of the IC of FIG. **1**.
(28) As shown in the cross-sectional view **200** of FIG. **2**, a dielectric layer **202** is disposed below the low-cost SOI structure **102**. The metal layer **108** is disposed vertically between the dielectric layer **202** and the insulating layer **106**. In some embodiments, the dielectric layer **202** contacts (e.g., directly contacts) the metal layer **108**. In some embodiments, the dielectric layer **202** is or comprises, for example, a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), a high-k dielectric material (e.g., a dielectric material with a dielectric constant greater than about 3.9, such as, hafnium oxide (HfO), tantalum oxide (TaO), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), aluminum oxide (AlO), zirconium oxide (ZrO), or the like), an oxide (e.g., SiO.sub.2), a nitride (e.g., SiN), an oxy-nitride (e.g., SiON), undoped silicate glass (USG), doped silicon dioxide (e.g., carbon doped silicon dioxide), borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), a spin-on glass (SOG), some other dielectric material, or a combination of the foregoing.
(29) In some embodiments, the dielectric layer **202** is or comprises a same material as the insulating layer **106**. For example, the dielectric layer **202** and the insulating layer **106** are borophosphosilicate glass (BPSG). In further embodiments, the insulating layer **106** and the ILD structure **118** are or comprise a same material. For example, the insulating layer **106** and the ILD structure **118** are borophosphosilicate glass (BPSG).
(30) A plurality of first isolation structures **204** are disposed in the device layer **104**. In some embodiments, the first isolation structures **204** extend into the device layer **104** from the front-side **102*f*** of the low-cost SOI structure **102**. The first isolation structures **204** may have angled sidewalls. In other embodiments, the sidewalls of the first isolation structures **204** may be substantially straight (e.g., vertical). The first isolation structures **204** extend vertically through the device layer **104** to the insulating layer **106**. The first isolation structures **204** have lower surfaces

(e.g., lowermost surfaces) that contact (e.g., directly contact) the insulating layer **106**. In some embodiments, the first isolation structures **204** may be or comprise, for example, an oxide (e.g., SiO2), a nitride (e.g., SiN), an oxy-nitride (e.g., SiON), a carbide (e.g., silicon carbide (SiC)), some other dielectric material, or a combination of the foregoing.
(31) A plurality of second isolation structures **206** are disposed in metal layer **108** and the dielectric layer **202**. The second isolation structures **206** extend vertically through the dielectric layer **202** and vertically through the metal layer **108** to the insulating layer **106**. The second isolation structures **206** have upper surfaces (e.g., uppermost surfaces) that contact (e.g., directly contact) the insulating layer **106**. The second isolation structures **206** and the first isolation structures **204** are laterally aligned (e.g., centered laterally), respectively. In some embodiments, the second isolation structures **206** may be or comprise, for example, an oxide (e.g., SiO2), a nitride (e.g., SiN), an oxy-nitride (e.g., SiON), a carbide (e.g., SiC), some other dielectric material, or a combination of the foregoing.
(32) A plurality of through-substrate vias (TSVs) **208***a-d* extend vertically through the device layer **104**. For example, the plurality of TSVs **208***a-d* comprise a first TSV **208***a*, a second TSV **208***b*, a third TSV **208***c*, and a fourth TSV **208***d* that each extend vertically through the device layer **104**. The TSVs **208***a-d* also extend vertically through the first isolation structures **204**, respectively. In other words, the TSVs **208***a-d* extend through the device layer **104** by extending vertically through the first isolation structures **204**. In some embodiments, the TSVs **208***a-d* may be or comprise, for example, copper (Cu), aluminum (Al), gold (Au), silver (Ag), platinum (Pt), or the like. In further embodiments, the TSVs **208***a-d* are referred to as back-side TSVs (BTSV).
(33) The first isolation structures **204** laterally surround the TSVs **208***a-d*, respectively. The first isolation structures **204** are configured to electrically isolate the TSVs **208***a-d* from the device layer **104**. For example, a first one of the first isolation structures **204** laterally surrounds the first TSV **208***a* and is configured to electrically isolate the first TSV **208***a* from the device layer **104**. The TSVs **208***a-d* also extend vertically through the insulating layer **106**.
(34) In some embodiments, the TSVs **208***a-d* also extend vertically through the metal layer **108**. In further embodiments, the TSVs **208***a-d* extend vertically through the second isolation structures **206**, respectively. In other words, the TSVs **208***a-d* extend through the metal layer **108** by extending vertically through the second isolation structures **206**.
(35) The second isolation structures **206** laterally surround the TSVs **208***a-d*, respectively. The second isolation structures **206** are configured to electrically isolate the TSVs **208***a-d* from the metal layer **108**. For example, a first one of the second isolation structures **206** laterally surrounds the first TSV **208***a* and is configured to electrically isolate the first TSV **208***a* from the metal layer **108**. In some embodiments, the TSVs **208***a-d* also extend vertically through the dielectric layer **202**. In further embodiments, the TSVs **208***a-d* extend through the dielectric layer **202** by extending vertically through the second isolation structures **206**.
(36) The TSVs **208***a-d* also extend vertically through a portion of the ILD structure **118**. The TSVs **208***a-d* extend vertically through the portion of the ILD structure **118** to corresponding conductive features of the interconnect structure **120**. The TSVs **208***a-d* are electrically coupled to the corresponding conductive features of the interconnect structure **120**, respectively. For example, the first TSV **208***a* extends vertically through the portion of the ILD structure **118** to a first conductive line **124***a* of the first plurality of conductive lines **124**, the second TSV **208***b* extends vertically through the portion of the ILD structure **118** to a second conductive line **124***b* of the first plurality of conductive lines **124**, the third TSV **208***c* extends vertically through the portion of the ILD structure **118** to a third conductive line **124***c* of the first plurality of conductive lines **124**, and the fourth TSV **208***d* extends vertically through the portion of the ILD structure **118** to a fourth conductive line **124***d* of the first plurality of conductive lines **124**.
(37) A second plurality of conductive lines **210***a-d* (e.g., metal lines) are disposed below the low-cost SOI structure **102** and the dielectric layer **202**. The second plurality of conductive lines **210***a-d*

are, at least partially, vertically separated from the metal layer **108** by the dielectric layer **202**. The TSVs **208***a-d* extend vertically from the corresponding conductive features of the interconnect structure **120** to corresponding conductive lines of the second plurality of conductive lines **210***a-d*. The TSVs **208***a-d* are electrically coupled to their corresponding conductive line of the second plurality of conductive lines **210***a-d*. For example, the first TSV **208***a* extends vertically from the first conductive line **124***a* of the first plurality of conductive lines **124** to a first conductive line **210***a* of the second plurality of conductive lines **210**, the second TSV **208***b* extends vertically from the second conductive line **124***b* of the first plurality of conductive lines **124** to a second conductive line **210***b* of the second plurality of conductive lines **210**, the third TSV **208***c* extends vertically from the third conductive line **124***c* of the first plurality of conductive lines **124** to a third conductive line **210***c* of the second plurality of conductive lines **210**, and the fourth TSV **208***d* extends vertically from the fourth conductive line **124***d* of the first plurality of conductive lines **124** to a fourth conductive line **210***d* of the second plurality of conductive lines **210**.
(38) Also shown in the cross-sectional view **200** of FIG. **2**, the TSVs **208***a-d* electrically couple their corresponding conductive feature of the interconnect structure **120** to their corresponding conductive line of the second plurality of conductive lines **210***a-d*. For example, the first TSV **208***a* electrically couples the first conductive line **124***a* of the first plurality of conductive lines **124** to the first conductive line **210***a* of the second plurality of conductive lines **210**, the second TSV **208***b* electrically couples the second conductive line **124***b* of the first plurality of conductive lines **124** to the second conductive line **210***b* of the second plurality of conductive lines **210**, the third TSV **208***c* electrically couples the third conductive line **124***c* of the first plurality of conductive lines **124** to the third conductive line **210***c* of the second plurality of conductive lines **210**, and the fourth TSV **208***d* electrically couples the fourth conductive line **124***d* of the first plurality of conductive lines **124** to the fourth conductive line **210***d* of the second plurality of conductive lines **210**. As such, corresponding voltages (or currents) may be applied to the TSVs **208***a-d* to specific features of the IC by applying the corresponding voltages to the second plurality of conductive lines **210***a-d*, and/or vice versa.
(39) For example, as shown in the cross-sectional view **200** of FIG. **2**, the second conductive line **210***b* of the second plurality of conductive lines **210** may be electrically coupled to the gate electrode **116** of the semiconductor device **110** (e.g., via the second TSV **208***b*, the second conductive line **124***b*, etc.). Thus, a first voltage (or current) may be applied to the gate electrode **116** of the semiconductor device **110** by applying the first voltage (or current) to the second conductive line **210***b* of the second plurality of conductive lines **210**. Further, the source/drain regions of the pair of source/drain regions **112** may be electrically coupled to corresponding ones of the second plurality of conductive lines **210**. For example, as shown in the cross-sectional view **200** of FIG. **2**, the fourth conductive line **210***d* of the second plurality of conductive lines **210** may be electrically coupled to one of the source/drain regions of the plurality of source/drain regions **112** (e.g., via the fourth TSV **208***d*, the fourth conductive line **124***d*, etc.). Thus, a second voltage (or current) may be applied to the one of the source/drain regions of the plurality of source/drain regions **112** by applying the second voltage (or current) to the fourth conductive line **210***d* of the second plurality of conductive lines **210**, and/or vice versa. Moreover, as shown in the cross-sectional view **200** of FIG. **2**, the third conductive line **210***c* of the second plurality of conductive lines **210** may be electrically coupled to the third conductive line **124***c* of the first plurality of conductive lines **124** (e.g., via the third TSV **208***c*). Thus, a third voltage (or current) may be applied to the third conductive line **124***c* of the first plurality of conductive lines **124** by applying the third voltage (or current) to the third conductive line **210***c* of the second plurality of conductive lines **210**, and/or vice versa.
(40) In some embodiments, the second plurality of conductive lines **210***a-d* may be or comprise, for example, copper (Cu), aluminum (Al), gold (Au), silver (Ag), platinum (Pt), or the like. In further embodiments, the second plurality of conductive lines **210***a-d* and the TSVs **208***a-d* may be or

comprise a same material. For example, the second plurality of conductive lines **210***a-d* and the TSVs **208***a-d* may be copper (Cu). In other embodiments, the second plurality of conductive lines **210***a-d* and the TSVs **208***a-d* may be or comprise a different material. In further embodiments, the second plurality of conductive lines **210***a-d* are referred to as a redistribution layer (RDL).
(41) The dielectric layer **202** is configured to electrically isolate the second plurality of conductive lines **210***a-d* from the metal layer **108**. In some embodiments, one or more of the second plurality of conductive lines **210** extend through the dielectric layer **202** to the metal layer **108**, such that the one or more of the second plurality of conductive lines **210** are electrically coupled to the metal layer **108**. Thus, by applying a voltage (or current) to the one or more of the second plurality of conductive lines **210**, the metal layer **108** may be biased.
(42) For example, as shown in the cross-sectional view **200** of FIG. **2**, the third conductive line **210***c* has a lateral portion and a vertical portion. The lateral portion of the third conductive line **210***c* contacts (e.g., directly contacts) the third TSV **208***c*. The lateral portion of the third conductive line **210***c* extends laterally below the dielectric layer **202** from the third TSV **208***c* to a first location. At the first location, the vertical portion of the third conductive line **210***c* extends vertically from the lateral portion of the third conductive line **210***c* through the dielectric layer **202** and to the metal layer **108**, such that the third conductive line **210***c* is electrically coupled to the metal layer **108**. Thus, the third voltage (or current) may be applied to the metal layer **108** by applying the third voltage (or current) to the third conductive line **210***c*, and/or a fourth voltage (or current) may be applied to the metal layer **108** by applying the fourth voltage (or current) to the third conductive line **124***c*. As such, the metal layer **108** may be biased, thereby improving the performance of the semiconductor device **110** (e.g., increased switching speed, reduced leakage current, etc.).
(43) FIG. **3** illustrates a cross-sectional view **300** of some other embodiments of the IC of FIG. **2**.
(44) As shown in the cross-sectional view **300** of FIG. **3**, a plurality of third isolation structures **302** are disposed in the device layer **104**. The third isolation structures **302** and the first isolation structures **204** are laterally aligned (e.g., centered laterally), respectively. The third isolation structures **302** may have angled sidewalls. In other embodiments, the sidewalls of the third isolation structures **302** may be substantially straight (e.g., vertical). In some embodiments, the third isolation structures **302** may be or comprise, for example, an oxide (e.g., SiO2), a nitride (e.g., SiN), an oxy-nitride (e.g., SiON), a carbide (e.g., SiC), some other dielectric material, or a combination of the foregoing.
(45) The third isolation structures **302** extends into the device layer **104** from the front-side **102***f* of the low-cost SOI structure **102**. The third isolation structures **302** extend partially through the device layer **104**. The first isolation structures **204** contact (e.g., directly contacts) the third isolation structures **302**, respectively. The first isolation structures **204** extend vertically from the third isolation structures **302**, respectively, to the insulating layer **106**. The TSVs **208***a-d* extend vertically through the third isolation structures **302**, respectively, and the first isolation structures **204**, respectively. In other words, the TSVs **208***a-d* extend through the device layer **104** by extending vertically through both the third isolation structures **302** and the first isolation structures **204**.
(46) The third isolation structures **302** laterally surround the TSVs **208***a-d*, respectively. The third isolation structures **302** are configured to electrically isolate portions of the TSVs **208***a-d* from the device layer **104**, and the first isolation structures **204** are configured to electrically isolate remaining portions of the TSVs **208***a-d* from the device layer **104**. For example, a first one of the third isolation structures **302** laterally surrounds an upper portion of the first TSV **208***a* and is configured to electrically isolate the upper portion of the first TSV **208***a* from the device layer **104**, and a first one of the first isolation structures **204** laterally surrounds a lower portion of the first TSV **208***a* and is configured to electrically isolate the lower portion of the first TSV **208***a* from the device layer **104**. In some embodiments, the third isolation structures **302** are referred to as shallow

trench isolation (STI) structures. In further embodiments, the first isolation structures **204** are referred to as deep trench isolation (DTI) structures.
(47) FIGS. **4**A-**4**D illustrate bottom views **400***a-d* of some embodiments of the IC of FIG. **3**. More specifically, FIG. **4**A illustrates a bottom view **400***a* of the device layer **104** of an embodiment of the IC of FIG. **3**, FIG. **4**B illustrates a bottom view **400***b* of the insulating layer **106** of the embodiment of the IC of FIG. **3**, FIG. **4**C illustrates a bottom view **400***c* of the metal layer **108** of the embodiment of the IC of FIG. **3**, and FIG. **4**D illustrates a bottom view **400***d* of the embodiment of the IC of FIG. **3**.
(48) As shown in the bottom views **400***a-d* of FIGS. **4**A-**4**D, the device layer **104** has a plurality of sidewalls **104***s*.sub.1-4. For example, the plurality of sidewalls **104***s*.sub.1-4 of the device layer **104** comprises a first sidewall **104***s*.sub.1 of the device layer **104**, a second sidewall **104***s*.sub.2 of the device layer **104**, a third sidewall **104***s*.sub.3 of the device layer **104**, and a fourth sidewall **104***s*.sub.4 of the device layer **104**. In some embodiments, the plurality of sidewalls **104***s*.sub.1-4 of the device layer **104** are outermost sidewalls of the device layer **104**.
(49) Further, the insulating layer **106** has a plurality of sidewalls **106***s*.sub.1-4. For example, the plurality of sidewalls **106***s*.sub.1-4 of the insulating layer **106** comprises a first sidewall **106***s*.sub.1 of the insulating layer **106**, a second sidewall **106***s*.sub.2 of the insulating layer **106**, a third sidewall **106***s*.sub.3 of the insulating layer **106**, and a fourth sidewall **106***s*.sub.4 of the insulating layer **106**. In some embodiments, the plurality of sidewalls **106***s*.sub.1-4 of the insulating layer **106** are outermost sidewalls of the insulating layer **106**.
(50) In addition, the metal layer **108** has a plurality of sidewalls **108***s*.sub.1-4. For example, the plurality of sidewalls **108***s*.sub.1-4 of the metal layer **108** comprises a first sidewall **108***s*.sub.1 of the metal layer **108**, a second sidewall **108***s*.sub.2 of the metal layer **108**, a third sidewall **108***s*.sub.3 of the metal layer **108**, and a fourth sidewall **108***s*.sub.4 of the metal layer **108**. In some embodiments, the plurality of sidewalls **108***s*.sub.1-4 of the metal layer **108** are outermost sidewalls of the metal layer **108**.
(51) Moreover, the dielectric layer **202** has a plurality of sidewalls **202***s*.sub.1-4. For example, the plurality of sidewalls **202***s*.sub.1-4 of the dielectric layer **202** comprises a first sidewall **202***s*.sub.1 of the dielectric layer **202**, a second sidewall **202***s*.sub.2 of the dielectric layer **202**, a third sidewall **202***s*.sub.3 of the dielectric layer **202**, and a fourth sidewall **202***s*.sub.4 of the dielectric layer **202**. In some embodiments, the plurality of sidewalls **202***s*.sub.1-4 of the dielectric layer **202** are outermost sidewalls of the dielectric layer **202**.
(52) In some embodiments, the plurality of sidewalls **104***s*.sub.1-4 of the device layer **104**, the plurality of sidewalls **106***s*.sub.1-4 of the insulating layer **106**, the plurality of sidewalls **108***s*.sub.1-4 of the metal layer **108**, and the plurality of sidewalls **202***s*.sub.1-4 of the dielectric layer **202** are substantially aligned (e.g., flush). For example, the first sidewall **104***s*.sub.1 of the device layer **104**, the first sidewall **106***s*.sub.1 of the insulating layer **106**, the first sidewall **108***s*.sub.1 of the metal layer **108**, and the first sidewall **202***s*.sub.1 of the dielectric layer **202** are substantially aligned (e.g., flush) with one another; the second sidewall **104***s*.sub.2 of the device layer **104**, the second sidewall **106***s*.sub.2 of the insulating layer **106**, the second sidewall **108***s*.sub.2 of the metal layer **108**, and the second sidewall **202***s*.sub.2 of the dielectric layer **202** are substantially aligned (e.g., flush) with one another; and so forth. In further embodiments, a footprint of the device layer **104**, a footprint of the insulating layer **106**, a footprint of the metal layer **108**, and a footprint of the dielectric layer **202** are substantially the same (e.g., a same width, length, and shape) as one another.
(53) Also shown in the bottom views **400***a-d* of FIGS. **4**A-**4**D, the conductive lines of the second plurality of conductive lines **210** may be laterally spaced from one another. As such, different voltages (or currents) may be applied to the different conductive lines of the second plurality of conductive lines **210** to selectively apply corresponding voltages (or currents) to specific features of the IC. Although not shown, it will be appreciated that a plurality of input/output (I/O) structures

(e.g., bond pads, solder bumps, etc.) may be disposed below (or in plane) with and electrically coupled to the second plurality of conductive lines **210**, respectively. In such embodiments, the corresponding voltages (or currents) may be applied to the specific features of the IC via the I/O structures, and/or vice versa.

(54) FIG. **5** illustrates a cross-sectional view **500** of some other embodiments of the IC of FIG. **3**.

(55) As shown in the cross-sectional view **500** of FIG. **5**, a first well region **502** is disposed in the device layer **104**. The first well region **502** is a region of the device layer **104** having a second doping type (e.g., p-type). A second well region **504** is also disposed in the device layer **104**. The second well region **504** is a region of the device layer **104** having the first doping type (e.g., n-type). A third well region **506** is also disposed in the device layer **104**. The third well region **506** is a region of the device layer **104** having the second doping type. The second well region **504** is disposed laterally between the first well region **502** and the third well region **506**.

(56) In some embodiments, a plurality of semiconductor devices **508***a-b* are disposed on/over the device layer **104**. For example, a first semiconductor device **508***a* (e.g., a first double-diffused MOSFET (DMOS)) and a second semiconductor device **508***b* (e.g., a second DMOS) are disposed on/over the device layer **104**. In some embodiments, the first semiconductor device **508***a* and the second semiconductor device **508***b* share a common drain region **510**. In other embodiments, the first semiconductor device **508***a* and the second semiconductor device **508***b* have their own drain regions that are laterally spaced. The drain region **510** is a region of the device layer **104** having the first doping type. The drain region **510** is disposed in the second well region **504**. A first one of the plurality of conductive contacts **122** is electrically coupled to the drain region **510**.

(57) The first semiconductor device **508***a* also comprises a source region **512**, a body contact region **514**, a gate dielectric **516**, and a gate electrode **518**. The source region **512** is a region of the device layer **104** having the first doping type. The source region **512** is disposed in the first well region **502**. The body contact region **514** is a region of the device layer **104** having the second doping type. The body contact region **514** is disposed in the first well region **502**. A second one of the plurality of conductive contacts **122** is electrically coupled to both the source region **512** and the body contact region **514**. The gate dielectric **516** is disposed over the device layer **104** and laterally between the source region **512** and the drain region **510**. The gate electrode **518** overlies the gate dielectric **516**. In some embodiments, the gate dielectric **516** partially overlies one of the third isolation structures **302**, which is disposed laterally between the source region **512** and the drain region **510**. A third one of the plurality of conductive contacts **122** is electrically coupled to the gate electrode **518**.

(58) The second semiconductor device **508***b* also comprises a source region **520**, a body contact region **522**, a gate dielectric **524**, and a gate electrode **526**. The source region **520** is a region of the device layer **104** having the first doping type. The source region **520** is disposed in the third well region **506**. The body contact region **522** is a region of the device layer **104** having the second doping type. The body contact region **522** is disposed in the third well region **506**. A fourth one of the plurality of conductive contacts **122** is electrically coupled to both the source region **520** and the body contact region **522**. The gate dielectric **524** is disposed over the device layer **104** and laterally between the source region **520** and the drain region **510**. The gate electrode **526** overlies the gate dielectric **524**. In some embodiments, the gate dielectric **524** partially overlies another one of the third isolation structures **302**, which is disposed laterally between the source region **520** and the drain region **510**. A fifth one of the plurality of conductive contacts **122** is electrically coupled to the gate electrode **526**.

(59) The first plurality of conductive lines **124** are disposed in a plurality of conductive layers **528***a-e* (e.g., metal layers). Each of the plurality of conductive layers **528***a-e* extend laterally through the ILD structure **118**. Each of the plurality of conductive layers **528***a-e* comprise a group of one or more of the first plurality of conductive lines **124**. The plurality of conductive layers **528***a-e* are disposed over one another. The plurality of conductive vias **126** extend vertically

between the plurality of conductive layers **528**_a-e_ and electrically couple the first plurality of conductive lines **124** of the plurality of conductive layers **528**_a-e_ together in a predefined manner.
(60) For example, the plurality of conductive layers **528**_a-e_ comprises a first conductive layer **528**_a_ (e.g., metal layer 1), a second conductive layer **528**_b_ (e.g., metal layer 2), a third conductive layer **528**_c_ (e.g., metal layer 3), a fourth conductive layer **528**_d_ (e.g., metal layer 4), and a fifth conductive layer **528**_e_ (e.g., metal layer 5) disposed in the ILD structure **118**. The first conductive layer **528**_a_ comprises a first group of conductive lines of the first plurality of conductive lines **124**, the second conductive layer **528**_b_ comprises a second group of conductive lines of the first plurality of conductive lines **124**, the third conductive layer **528**_c_ comprises a third group of conductive lines of the first plurality of conductive lines **124**, the fourth conductive layer **528**_d_ comprises a fourth group of conductive lines of the first plurality of conductive lines **124**, and the fifth conductive layer **528**_e_ comprises a fifth group of conductive lines of the first plurality of conductive lines **124**. The second conductive layer **528**_b_ is disposed over the first conductive layer **528**_a_, the third conductive layer **528**_c_ is disposed over the second conductive layer **528**_b_, the fourth conductive layer **528**_d_ is disposed over the third conductive layer **528**_c_, and the fifth conductive layer **528**_e_ is disposed over the fourth conductive layer **528**_d_. In some embodiments, the first conductive layer **528**_a_ is disposed nearer the device layer **104** than any other of the plurality of conductive layers **528**_a-e_. In further embodiments, the TSVs **208** are electrically coupled to corresponding conductive lines of the first conductive layer **528**_a_. It will be appreciated that the plurality of conductive layers **528**_a-e_ is not limited to only five conductive layers, but rather the plurality of conductive layers **528**_a-e_ may comprise any suitable number of conductive layers.
(61) FIG. **6** illustrates a cross-sectional view **600** of some other embodiments of the IC of FIG. **5**.
(62) As shown in the cross-sectional view **600** of FIG. **6**, the low-cost SOI structure **102** comprises a conductive layer **602**. The device layer **104** is disposed over the insulating layer **106** and the conductive layer **602**. The insulating layer **106** is disposed vertically between the conductive layer **602** and the device layer **104**. The insulating layer **106** vertically separates the conductive layer **602** from the device layer **104**. In some embodiments, the conductive layer **602** is utilized in place of the metal layer **108** (e.g., is disposed in the IC in a substantially similar manner as the metal layer **108**).
(63) The conductive layer **602** is disposed below both the insulating layer **106** and the device layer **104**. The conductive layer **602** is or comprises a conductive material. For example, in some embodiments, the conductive layer **602** is or comprises a doped semiconductor material (e.g., doped silicon (Si), doped germanium (Ge), etc.), a metal (e.g., Cu, Al, W, Au, Ag, Pt, etc.), a non-metal conductive material, some other conductive material, or a combination of the foregoing. In further embodiments, the conductive layer **602** is a doped semiconductor material having a doping concentration of a type of dopants (e.g., n-type dopants or p-type dopants) that is greater than or equal to about 1×10.sup.15 atoms per cubic centimeter (atoms/cm.sup.3). In yet further embodiments, the conductive layer **602** may further reduce a cost to fabricate the IC.
(64) FIGS. **7-15** illustrate a series of cross-sectional views **700-1500** of some embodiments of a method for forming an integrated chip (IC) comprising a low-cost SOI structure **102**. Although FIGS. **7-15** are described with reference to a method, it will be appreciated that the structures shown in FIGS. **7-15** are not limited to the method but rather may stand alone separate of the method.
(65) As shown in the cross-sectional view **700** of FIG. **7**, a workpiece **702** is received. The workpiece **702** comprises a device substrate **704**. The device substrate **704** has a front-side **704**_f_ and a back-side **704**_b_ opposite the front-side **704**_f_. The semiconductor material may be or comprise, for example, silicon (Si), germanium (Ge), silicon-germanium (SiGe), gallium arsenide (GaAs), some other semiconductor material, or a combination of the foregoing. In some embodiments, the device substrate **704** is silicon (Si). In further embodiments, the device substrate **704** is monocrystalline silicon. In some embodiments, the device substrate **704** has a thickness greater than about 15

micrometers (μm). In further embodiments, the thickness of the device substrate **704** is about 750 μm. The device substrate **704** is a bulk semiconductor material.
(66) A plurality of first isolation structures **204** are formed in the device substrate **704**. A plurality of third isolation structures **302** are also formed in the device substrate **704**. Further, a first well region **502**, a second well region **504**, and a third well region **506** are formed in the device substrate **704**. A plurality of semiconductor devices **508***a-b* are formed on/over the device substrate **704**. For example, a first semiconductor device **508***a* and a second semiconductor device **508***b* are formed on/over the device substrate **704**. In some embodiments, the first semiconductor device **508***a* and the second semiconductor device **508***b* share a common drain region **510**. The first semiconductor device **508***a* also comprises a source region **512**, a body contact region **514**, a gate dielectric **516**, and a gate electrode **518**. The second semiconductor device **508***b* also comprises a source region **520**, a body contact region **522**, a gate dielectric **524**, and a gate electrode **526**.
(67) An interlayer dielectric (ILD) structure **118** is formed over both the plurality of semiconductor devices **508***a-b* and the device substrate **704**. An interconnect structure **120** is formed in the ILD structure **118** and over the device substrate **704**. The interconnect structure **120** comprises a plurality of conductive contacts **122**, a first plurality of conductive lines **124**, and a plurality of conductive vias **126**. The first plurality of conductive lines **124** are disposed in a plurality of conductive layers **528***a-e*. For example, the plurality of conductive layers **528***a-e* comprises a first conductive layer **528***a*, a second conductive layer **528***b*, a third conductive layer **528***c*, a fourth conductive layer **528***d*, and a fifth conductive layer **528***e* disposed in the ILD structure **118**. The plurality of first isolation structures **204**, the plurality of third isolation structures **302**, the first well region **502**, the second well region **504**, the third well region **506**, the plurality of semiconductor devices **508***a-b*, the ILD structure **118**, and the interconnect structure **120** may be formed by known complementary metal-oxide-semiconductor (CMOS) processes.
(68) As shown in the cross-sectional view **800** of FIG. **8**, the device substrate **704** (see, FIG. **7**) is thinned to form a device layer **104**. The device substrate **704** is thinned by performing a thinning process **802** on the back-side **704***b* (see, FIG. **7**) of the device substrate **704** to reduce the thickness of the low-cost SOI structure **102** from a first thickness T.sub.1 to a second thickness T.sub.2. The first thickness T.sub.1 is greater than about 15 μm. In some embodiments, the first thickness T.sub.1 is about 750 μm. In further embodiments, the second thickness T.sub.2 is between about 2 μm and about 15 μm. The device layer **104** corresponds to a portion of the device substrate **704** that remains after the thinning process **802**. The device layer **104** has a front-side **104***f* and a back-side **104***b*. In some embodiments, the thinning process **802** may be or comprise, for example, a chemical mechanical polishing (CMP) process, a mechanical grinding process, an etching process, some other thinning process, or a combination of the foregoing. The dashed line in the cross-sectional view **800** of FIG. **8** illustrates the portion of the device substrate **704** that is removed by the thinning process **802**.
(69) As shown in the cross-sectional view **900** of FIG. **9**, an insulating layer **106** is formed over/on the back-side **104***b* of the device layer **104**. The insulating layer **106** may be formed by, for example, chemical vapor deposition (CVD), high density plasma CVD (HDPCVD), high aspect ratio process (HARP), physical vapor deposition (PVD), atomic layer deposition (ALD), a spin-on process, some other deposition process, or a combination of the foregoing. In some embodiments, the insulating layer **106** is formed via a blanket deposition process.
(70) Also shown in the cross-sectional view **900** of FIG. **9**, a metal layer **108** is formed over/on the insulating layer **106**, such that the insulating layer **106** vertically separates the metal layer **108** from the device layer **104**. The metal layer **108** may be formed by, for example, CVD, HDPCVD, PVD, ALD, sputtering, electrochemical plating, electroless plating, some other deposition process, or a combination of the foregoing. In some embodiments, the metal layer **108** is formed via a blanket deposition process. In some embodiments, after the metal layer **108** is formed, formation of a low-cost SOI structure **102** is complete. The low-cost SOI structure **102** has a front-side **102***f* and a

back-side **102***b* opposite the front-side **102***f*. The front-side **102***f* of the low-cost SOI structure **102** corresponded to the front-side **104***f* (see, FIG. **8**) of the device layer **104**.

(71) It will be appreciated that, in some embodiments, the conductive layer **602** may be formed in place of the metal layer **108**. More specifically, in some embodiments, the conductive layer **602** is a doped semiconductor material and is formed in place of the metal layer **108**. The conductive layer **602** may be formed by, for example, CVD, HDPCVD, PVD, ALD, sputtering, an epitaxial process, some other deposition or growth process, or a combination of the foregoing. In some embodiments, by forming the conductive layer **602**, a cost to fabricate the IC may be further reduced (e.g., more so than forming the IC in the described manner with the metal layer **108**). More specifically, in some embodiments, it may cost less to form the low-cost SOI structure **102** due to a cost to form the doped semiconductor material being less than a cost to form the metal layer **108**.

(72) Also shown in the cross-sectional view **900** of FIG. **9**, a dielectric layer **202** is formed over/on the metal layer **108**, such that the metal layer **108** vertically separates the dielectric layer **202** from the insulating layer **106**. The dielectric layer **202** may be formed by, for example, CVD, HDPCVD, HARP, PVD, ALD, a spin-on process, some other deposition process, or a combination of the foregoing. In some embodiments, the dielectric layer **202** is formed via a blanket deposition process.

(73) As shown in the cross-sectional view **1000** of FIG. **10**, a plurality of first openings **1002** are formed in low-cost SOI structure **102**. The first openings **1002** extend vertically into the low-cost SOI structure **102** from the back-side **102***b* of the low-cost SOI structure **102**. The first openings **1002** extend vertically through both the dielectric layer **202** and the metal layer **108**. The first openings **1002** extend vertically to the insulating layer **106**. The first openings **1002** expose portions of the insulating layer **106**, respectively. In some embodiments, the first openings **1002** and the first isolation structures **204** are laterally aligned (e.g., centered laterally), respectively.

(74) In some embodiments, a process for forming the first openings **1002** comprises forming a patterned masking layer (not shown) (e.g., positive/negative photoresist, a hardmask, etc.) over the dielectric layer **202**. The patterned masking layer may be formed by forming a masking layer (not shown) on the dielectric layer **202** (e.g., via a spin-on process), exposing the masking layer to a pattern (e.g., via a lithography process, such as photolithography, extreme ultraviolet lithography, or the like), and developing the masking layer to form the patterned masking layer. Thereafter, with the patterned masking layer in place, an etching process is performed on the dielectric layer **202** and the metal layer **108** to selectively etch the dielectric layer **202** and the metal layer **108** according to the patterned masking layer. The etching process removes unmasked portions of the dielectric layer **202** and unmasked portions of the metal layer **108**, thereby forming the first openings **1002**. In some embodiments, the etching process may be or comprise, for example, a wet etching process, a dry etching process, a reactive ion etching (RIE) process, some other etching process, or a combination of the foregoing. In further embodiments, the etching process stops on the insulating layer **106** (e.g., the insulating layer **106** acts as an etch stop layer during the etching process).

(75) As shown in the cross-sectional view **1100** of FIG. **11**, a plurality of second isolation structures **206** are formed in the plurality of first openings **1002** (see, FIG. **9**), respectively. The plurality of second isolation structures **206** extend vertically into the low-cost SOI structure **102** from the back-side **102***b* of the low-cost SOI structure **102**. The plurality of second isolation structures **206** extend vertically through both the dielectric layer **202** and the metal layer **108**. The plurality of second isolation structures **206** extend vertically to the insulating layer **106**. In some embodiments, the plurality of second isolation structures **206** and the first isolation structures **204** are laterally aligned (e.g., centered laterally), respectively.

(76) In some embodiments, a process for forming the plurality of second isolation structures **206** comprises filling the first openings **1002** with a dielectric material. In some embodiments, a process for filling the first openings **1002** with the dielectric material comprises depositing the

dielectric material on the dielectric layer **202** and in the first openings **1002**. The dielectric material may be or comprise, for example, an oxide (e.g., SiO2), a nitride (e.g., SiN), an oxy-nitride (e.g., SiON), a carbide (e.g., SiC), some other dielectric material, or a combination of the foregoing. The dielectric material may be deposited by, for example, CVD, PVD, ALD, some other deposition process, or a combination of the foregoing. In some embodiments, after the dielectric material is deposited, a planarization process (e.g., CMP) is performed on the dielectric material to remove an upper portion of the dielectric material, thereby leaving lower portions of the dielectric material in the first openings **1002** as the plurality of second isolation structures **206**. In further embodiments, the planarization process co-planarizes an upper surface of the dielectric layer **202** and upper surfaces of the plurality of second isolation structures **206**.
(77) As shown in the cross-sectional view **1200** of FIG. **12**, a plurality of second openings **1202** are formed in the dielectric layer **202**. The second openings **1202** extend vertically through the dielectric layer **202** to the metal layer **108**. The second openings **1202** expose portions of metal layer **108**, respectively.
(78) In some embodiments, a process for forming the second openings **1202** comprises forming a patterned masking layer (not shown) (e.g., positive/negative photoresist, a hardmask, etc.) over the dielectric layer **202** and the second isolation structures **206**. The patterned masking layer may be formed by forming a masking layer (not shown) on the dielectric layer **202** and on the second isolation structures **206** (e.g., via a spin-on process), exposing the masking layer to a pattern (e.g., via a lithography process, such as photolithography, extreme ultraviolet lithography, or the like), and developing the masking layer to form the patterned masking layer. Thereafter, with the patterned masking layer in place, an etching process is performed on the dielectric layer **202** to selectively etch the dielectric layer **202** according to the patterned masking layer. The etching process removes unmasked portions of the dielectric layer **202**, thereby forming the second openings **1202**. In some embodiments, the etching process may be or comprise, for example, a wet etching process, a dry etching process, a RIE process, some other etching process, or a combination of the foregoing. In further embodiments, the etching process stops on the metal layer **108** (e.g., the metal layer **108** acts as an etch stop layer during the etching process).
(79) As shown in the cross-sectional view **1300** of FIG. **13**, a plurality of third openings **1302** are formed in the dielectric layer **202**, the low-cost SOI structure **102**, and the ILD structure **118**. The third openings **1302** extend vertically into the dielectric layer **202**. The third openings **1302** also extend vertically into the low-cost SOI structure **102** from the back-side **102***b* of the low-cost SOI structure **102**. The third openings **1302** also extend vertically through a portion of the ILD structure **118** to corresponding conductive features of the interconnect structure **120**. For example, the third openings **1302** extend vertically through the portion of the ILD structure **118** to corresponding conductive lines of the first conductive layer **528***a*. The third openings **1302** expose a portion of their corresponding conductive line of the first conductive layer **528***a*. For example, one of the third openings **1302** extends vertically through the low-cost SOI structure **102** and the portion of the ILD structure **118** to one of the conductive lines of the first conductive layer **528***a*, thereby exposing a portion of the one of the conductive lines of the first conductive layer **528***a*.
(80) The third openings **1302** also extend vertically through portions of the insulating layer **106** that were exposed by forming the first openings **1002**. Further, the third openings **1302** extend vertically through the second isolation structures **206**, respectively. In other words, the third openings **1302** extend through the dielectric layer **202** and the metal layer **108** by extending vertically through the second isolation structures **206**. The second isolation structures **206** laterally surround the third openings **1302**, respectively. The third openings **1302** also extend vertically through the insulating layer **106**.
(81) In addition, the third openings **1302** extend vertically through the first isolation structures **204**, respectively. Moreover, the third openings **1302** may extend vertically through the third isolation structures **302**, respectively. In other words, the third openings **1302** extend through the device

layer **104** by extending vertically through the first isolation structures **204** and the third isolation structures **302**. The first isolation structures **204** laterally surround the third openings **1302**, respectively. The third isolation structures **302** laterally surround the third openings **1302**, respectively.

(82) In some embodiments, a process for forming the third openings **1302** comprises forming a patterned masking layer (not shown) (e.g., positive/negative photoresist, a hardmask, etc.) over the dielectric layer **202**, the second isolation structures **206**, and the second openings **1202**. The patterned masking layer may be formed by forming a masking layer (not shown) on the dielectric layer **202**, on the second isolation structures **206**, and in the second openings **1202** (e.g., via a spin-on process); exposing the masking layer to a pattern (e.g., via a lithography process, such as photolithography, extreme ultraviolet lithography, or the like); and developing the masking layer to form the patterned masking layer. Thereafter, with the patterned masking layer in place, an etching process is performed on the second isolation structures **206**, the insulating layer **106**, the first isolation structure **204**, the third isolation structures **302**, and a portion of the ILD structure **118** according to the patterned masking layer. The etching process removes unmasked portions of the second isolation structures **206**, unmasked portions of the insulating layer **106**, unmasked portions of the first isolation structure **204**, unmasked portions of the third isolation structures **302**, and unmasked portions of the portion of the ILD structure **118**, thereby forming the third openings **1302**. In some embodiments, the etching process may be or comprise, for example, a wet etching process, a dry etching process, a RIE process, some other etching process, or a combination of the foregoing. In further embodiments, the etching process stops on the corresponding conductive lines of the first conductive layer **528***a* (e.g., the corresponding conductive lines of the first conductive layer **528***a* act as etch stop structures during the etching process).

(83) As shown in the cross-sectional view **1400** of FIG. **14**, a conductive material **1402** is formed on the dielectric layer **202** and in the third openings **1302** (see, FIG. **13**). The conductive material **1402** is formed so that the conductive material **1402** fills the third openings **1302**. By filling the third openings **1302** with the conductive material **1402**, a plurality of through-substrate vias (TSVs) **208** are formed. In other words, the plurality of TSVs **208** are formed in the third openings **1302**, respectively. The plurality of TSVs **208** extend vertically through the dielectric layer **202**, the low-cost SOI structure **102**, and the portion of the ILD structure **118** to the corresponding conductive lines of the first conductive layer **528***a*. The plurality of TSVs **208** also extend through corresponding second isolation structures **206**, first isolation structure **204**, and third isolation structures **302**. The plurality of TSVs **208** are electrically coupled to their corresponding conductive line of the first conductive layer **528***a*.

(84) In some embodiments, the conductive material **1402** is or comprises, for example, copper (Cu), aluminum (Al), gold (Au), silver (Ag), platinum (Pt), polysilicon, some other conductive material, or a combination of the foregoing. In further embodiments, the conductive material **1402** is referred to as a metal material. In some embodiments, a process for forming the conductive material comprises depositing the conductive material **1402** on the dielectric layer **202** and in the third openings **1302**, such that the third openings **1302** are filled with the conductive material **1402**. The conductive material **1402** may be deposited by, for example, CVD, HDPCVD, PVD, ALD, sputtering, electrochemical plating, electroless plating, some other deposition process, or a combination of the foregoing.

(85) As shown in the cross-sectional view **1500** of FIG. **15**, a second plurality of conductive lines **210** (e.g., metal lines) are formed over/on the dielectric layer **202**, the TSVs **208**, and the second isolation structures **206**. The second plurality of conductive lines **210** are electrically coupled to the plurality of TSVs **208**, respectively. In some embodiments, one or more of the second plurality of conductive lines **210** are also electrically coupled to the metal layer **108**.

(86) In some embodiments, a process for forming the second plurality of conductive lines **210** comprises forming a patterned masking layer (not shown) (e.g., positive/negative photoresist, a

hardmask, etc.) over the conductive material **1402** (see, FIG. **14**). The patterned masking layer may be formed by forming a masking layer (not shown) on the conductive material **1402** (e.g., via a spin-on process), exposing the masking layer to a pattern (e.g., via a lithography process, such as photolithography, extreme ultraviolet lithography, or the like), and developing the masking layer to form the patterned masking layer. Thereafter, with the patterned masking layer in place, an etching process is performed on the conductive material **1402** to selectively etch the conductive material **1402** according to the patterned masking layer. The etching process removes unmasked portions of the conductive material **1402**, thereby forming the second plurality of conductive lines **210**. In some embodiments, the etching process may be or comprise, for example, a wet etching process, a dry etching process, a RIE process, some other etching process, or a combination of the foregoing. In further embodiments, the etching process stops on the dielectric layer **202** (e.g., the dielectric layer **202** acts as an etch stop layer during the etching process). It will be appreciated that the second plurality of conductive lines **210** (and/or the plurality of TSVs **208**) may be formed by other suitable processes (e.g., a damascene process). It will be appreciated that, after the second plurality of conductive lines **210** are formed, a plurality of input/output (I/O) structures (e.g., bond pads, solder bumps, etc.) may be formed below (or in plane) with and electrically coupled to the second plurality of conductive lines **210**, respectively.
(87) FIG. **16** illustrates a flowchart **1600** of some embodiments of a method for forming an integrated chip (IC) comprising a low-cost SOI structure. While the flowchart **1600** of FIG. **16** is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.
(88) At act **1602**, a workpiece comprising an interlayer dielectric (ILD) structure disposed over a first side of a device substrate is received. FIG. **7** illustrates a cross-sectional view **700** of some embodiments corresponding to act **1602**.
(89) At act **1604**, a device layer is formed by performing a thinning process on a second side of the device substrate opposite the first side of the device substrate. FIG. **8** illustrates a cross-sectional view **800** of some embodiments corresponding to act **1604**.
(90) At act **1606**, an insulating layer is formed on a side of the device layer, wherein the side of the device layer is opposite the ILD structure. FIG. **9** illustrates a cross-sectional view **900** of some embodiments corresponding to act **1606**.
(91) At act **1608**, a metal layer is formed over the insulating layer and the device layer. FIG. **9** illustrates a cross-sectional view **900** of some embodiments corresponding to act **1608**.
(92) At act **1610**, a dielectric layer is formed over the metal layer and the insulating layer. FIG. **9** illustrates a cross-sectional view **900** of some embodiments corresponding to act **1610**.
(93) At act **1612**, one or more through-substrate vias (TSVs) are formed extending through the dielectric layer, the metal layer, the insulating layer, the device layer, and a portion of the ILD structure. FIGS. **10-14** illustrate a series of cross-sectional views **1000-1400** of some embodiments corresponding to act **1612**.
(94) At act **1614**, a plurality of conductive lines are formed over the dielectric layer and electrically coupled to the one or more TSVs. FIG. **15** illustrates a cross-sectional view **1500** of some embodiments corresponding to act **1614**.
(95) In some embodiments, the present application provides an integrated chip (IC). The IC comprises a substrate. The substrate comprises a metal layer, a device layer disposed over the metal layer, and an insulating layer disposed vertically between the metal layer and the device layer. The device layer is a semiconductor material. A semiconductor device is disposed on the device layer. An interlayer dielectric (ILD) layer is disposed over the semiconductor device and the substrate.

(96) In some embodiments, the present application provides an integrated chip (IC). The IC comprises a substrate. The substrate comprises a conductive layer, a device layer disposed over the conductive layer, and an insulating layer disposed vertically between the conductive layer and the device layer. The device layer is a semiconductor material. A semiconductor device is disposed on the device layer. An interlayer dielectric (ILD) layer is disposed over the semiconductor device and the substrate. A conductive interconnect structure is embedded in the ILD structure and disposed over the substrate, wherein the conductive interconnect structure comprises a group of conductive lines that define a first layer of conducive lines. A first conductive structure extends vertically through the ILD structure, the device layer, and the insulating layer, wherein the first conductive structure electrically couples a first conductive line of the group of conductive lines to the conductive layer.
(97) In some embodiments, the present application provides a method for forming an integrated chip (IC). The method comprises receiving a workpiece, wherein the workpiece comprises a semiconductor device over a device substrate and comprises an interlayer dielectric (ILD) structure over the semiconductor device and over a first side of the device substrate, wherein the device substrate is a semiconductor material. A device layer is formed by performing a thinning process on a second side of the device substrate that reduces a thickness of the device substrate, wherein the second side of the device substrate is opposite the first side of the device substrate. After the thinning process, an insulating layer is formed on a side of the device layer, wherein the side of the device layer is opposite the ILD structure. A metal layer is formed on the insulating layer, such that the insulating layer vertically separates the metal layer from the device layer.
(98) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming an integrated chip (IC), the method comprising: receiving a workpiece, wherein the workpiece comprises a semiconductor device over a device substrate and comprises an interlayer dielectric (ILD) structure over the semiconductor device and over a first side of the device substrate, wherein the device substrate is a semiconductor material; forming a device layer by performing a thinning process on a second side of the device substrate that reduces a thickness of the device substrate, wherein the second side of the device substrate is opposite the first side of the device substrate; after the thinning process, forming an insulating layer on a side of the device layer, wherein the side of the device layer is opposite the ILD structure; and forming a metal layer on the insulating layer, such that the insulating layer vertically separates the metal layer from the device layer; forming a dielectric layer on the metal layer, such that the metal layer vertically separates the dielectric layer from the insulating layer; forming a first opening that extends vertically through the dielectric layer and vertically through the metal layer, wherein the first opening exposes a portion of the insulating layer; forming a first isolation structure in the first opening by filling the opening with a dielectric material; forming a second opening that extends vertically through the first isolation structure, the portion of the insulating layer, the device layer, and the ILD structure; and forming a first through-substrate via (TSV) in the second opening.
2. The method of claim 1, wherein a first side of the insulating layer directly contacts the metal layer and a second side of the insulating layer directly contacts the device layer, the second side

being opposite the first side.
3. The method of claim 1, wherein the ILD structure and the insulating layer comprise a same material.
4. The method of claim 1, wherein the device layer has a thickness ranging between about 2 micrometers (μm) and about 15 μm.
5. The method of claim 1, further comprising: forming a first metal wire within the dielectric layer, wherein a portion of the dielectric layer vertically separates the first metal wire from the metal layer.
6. The method of claim 5, wherein the first metal wire is electrically coupled to the first TSV.
7. The method of claim 6, wherein the first metal wire comprises: a lateral portion that extends laterally below the dielectric layer from the first TSV to a first location; and a vertical portion that extends from the first location vertically through the dielectric layer to the metal layer, such that the first metal wire electrically couples the first TSV to the metal layer.
8. The method of claim 7, further comprising: forming a second TSV extending through the device layer and the dielectric layer, wherein the second TSV is laterally spaced from the first TSV; and forming a second metal wire electrically coupled to the second TSV and disposed below the dielectric layer, wherein the dielectric layer vertically separates the second metal wire from the metal layer, and wherein the second metal wire is laterally spaced from the first metal wire.
9. The method of claim 1, further comprising: forming a second isolation structure in the device layer, wherein the first TSV extends through the second isolation structure, and wherein the second isolation structure electrically isolates the first TSV from the device layer.
10. The method of claim 9, wherein the first isolation structure has tapered outer sidewalls that define a first isolation structure width that varies along sidewalls of the first TSV.
11. The method of claim 10, wherein the second isolation structure has outer sidewalls whose sidewall angle differs from that of the tapered outer sidewalls of the first isolation structure.
12. The method of claim 9: wherein the first isolation structure has a bottom surface that terminates at an interface where a lower surface of the device layer meets an upper surface of the insulating layer; and wherein the second isolation structure has a top surface that terminates at an interface where a lower surface of the insulating layer meets an upper surface of the metal layer.
13. A method for forming an integrated chip (IC), the method comprising: receiving a workpiece, wherein the workpiece comprises a semiconductor device over a device substrate and comprises an interlayer dielectric (ILD) structure over the semiconductor device and over a first side of the device substrate, wherein the device substrate is a semiconductor material that includes silicon; forming a device layer by performing a thinning process on a second side of the device substrate that reduces a thickness of the device substrate, wherein the second side of the device substrate is opposite the first side of the device substrate; after the thinning process, forming an insulating layer on a side of the device layer, wherein the side of the device layer is opposite the ILD structure; and forming a metal layer on the insulating layer, such that the insulating layer vertically separates the metal layer from the device layer; forming a dielectric layer on the metal layer, such that the metal layer vertically separates the dielectric layer from the insulating layer; forming a first opening that extends vertically through the dielectric layer and vertically through the metal layer, wherein the first opening exposes a portion of the insulating layer; forming a first isolation structure in the first opening by filling the opening with a dielectric material; forming a second opening that extends vertically through the first isolation structure, the portion of the insulating layer, the device layer, and the ILD structure; and forming a first through-substrate via (TSV) in the second opening.
14. The method of claim 13, wherein the metal layer is a single continuous metal body having a thickness of between 1 micrometer and 5 micrometers.
15. The method of claim 13, wherein a first side of the insulating layer directly contacts the metal layer and a second side of the insulating layer directly contacts the device layer, the second side being opposite the first side.

16. The method of claim 13, wherein the device layer has a thickness ranging between about 2 micrometers (μm) and about 15 μm.

17. The method of claim 13, further comprising: forming a first metal wire prior to forming the dielectric layer, wherein the dielectric layer vertically separates the first metal wire from the metal layer.

18. The method of claim 13, further comprising: forming a second isolation structure in the device layer, wherein the second isolation structure electrically isolates the first TSV from the device layer.

19. The method of claim 18: wherein the first isolation structure has tapered outer sidewalls that are angled at a first angle relative to a lower surface of the device layer such that the first isolation structure has a first isolation structure width that varies as measured at different depths of the first TSV and as measured perpendicular to outer sidewalls of the first TSV; and wherein the second isolation structure is defined between vertical outer sidewalls that are angled at a second angle as measured relative to the lower surface of the device layer, wherein the second angle differs from the first angle.

20. A method for forming an integrated chip (IC), the method comprising: receiving a workpiece, wherein the workpiece comprises a semiconductor device over a device substrate and comprises an interlayer dielectric (ILD) structure over the semiconductor device and over a first side of the device substrate, wherein the device substrate is a semiconductor material; forming a device layer by performing a thinning process on a second side of the device substrate that reduces a thickness of the device substrate, wherein the second side of the device substrate is opposite the first side of the device substrate; after the thinning process, forming an insulating layer on a side of the device layer, wherein the side of the device layer is opposite the ILD structure; and forming a metal layer on the insulating layer, such that the insulating layer vertically separates the metal layer from the device layer; forming a dielectric layer on the metal layer, such that the metal layer vertically separates the dielectric layer from the insulating layer; forming a first opening that extends vertically through the dielectric layer and vertically through the metal layer, wherein the first opening exposes a portion of the insulating layer; forming an isolation structure in the first opening by filling the opening with a dielectric material; forming a second opening that extends vertically through the isolation structure, the portion of the insulating layer, the device layer, and the ILD structure; and forming a through-substrate via (TSV) comprising copper in the second opening.